Venkata Lakshmi Parimala Pasupuleti

Senior Software Engineer · AI-First Engineering · Agent-Driven Workflows · Real Estate Tech · Python · TypeScript · React · AWS · Observability · 9yr Production

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PROFESSIONAL SUMMARY

Senior software engineer with 9+ years of production experience and a specialization in AI-first engineering workflows, agent-driven product features, and real estate technology - the exact combination CoStar's greenfield AI platform requires. I use Claude Code as my primary engineering tool daily, building and shipping AI systems end-to-end with AI-augmented workflows that compress development cycles without sacrificing quality. AgentProbe demonstrates AI-first product engineering: I designed and shipped a production AI evaluation platform from scratch - agent orchestration with LangGraph, FastAPI backend with 15 endpoints, CI-integrated regression gate, structured tracing, and a 5-metric quality engine - using AI-assisted development throughout. LeaseI shows the agent-driven product pattern at scale: an 8-agent workflow automation system with 93.6% precision across 1,240 test cases, built in the real estate domain I know from 8 years at Entrata. That domain knowledge - property data, lease workflows, compliance requirements, the operational realities of managing properties at scale - is directly transferable to CoStar's commercial real estate platform. At Entrata I owned mission-critical Python backend systems for 7 years at 99.5% uptime, building with Python, TypeScript, React.js, PostgreSQL, RabbitMQ, Docker, AWS, and GitHub Actions CI/CD across 120+ Agile sprints. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026) - Virginia local.

TECHNICAL SKILLS

AI-First Engineering & Agent Workflows: Python (9yr), TypeScript, React.js, AI-First Engineering, Agent-Driven Workflows, LangGraph, LangChain, MCP, Multi-Agent Orchestration, Vibe Coding, Claude Code (daily), Generative AI, LLM Integration, Prompt Engineering, AI-Powered Product Features, AI Evaluation, Human-in-the-Loop, HITL Guardrails, Output Validation, AI Code Generation, AI Test Generation

Backend, APIs & Data: FastAPI, Flask, REST APIs, API Design, Async Processing, RabbitMQ, Microservices, Distributed Systems, PostgreSQL, MySQL, Oracle, Redis, SQL, NoSQL, High-Throughput Systems, Performance Optimization, Query Optimization, Indexing, Reliability Engineering, Idempotency, ChromaDB, Vector Databases, RAG Pipelines, Semantic Search

Cloud, Observability & Engineering Excellence: AWS (S3, EC2, RDS, Glue, Lambda), Azure-compatible, Docker, Microservices, CI/CD (GitHub Actions), Automated Testing, Unit Testing, Performance Testing, Observability, Structured Logging, Distributed Tracing, Monitoring, Alerting, Design Documents, Technical Mentoring, Code Reviews, Agile/Scrum, Real Estate Domain (8yr)

EXPERIENCE

Magizhs Tech Solution
May 2025 - Aug 2025

AI/ML Engineer - AI-First Product Delivery

- Led 0→1 delivery of a production AI-powered application (LangGraph, FastAPI, Python) using AI-first engineering workflows throughout - AI-augmented architecture design, code generation, test generation, and debugging; delivered production-ready system with evaluation pipeline, CI validation gate, and observability infrastructure in weeks as sole technical owner.

George Mason University
Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate

- Built ML pipelines (PyTorch, scikit-learn, R^2=0.977) on AWS cloud infrastructure; designed observability dashboards improving research comprehension 40%; delivered technical output across 3 research pipelines; mentored 600+ students in Python, SQL, and distributed systems - technical upskilling across diverse audiences.

Entrata - PropTech / Real Estate SaaS
Sep 2018 - Aug 2024

Senior Software Engineer · Lehi, UT (Remote)

- **Production software in the real estate domain** - owned mission-critical Python backend systems for 7 years at 99.5% uptime serving 100K+ residential units and 1M+ property records; built React.js + TypeScript frontend features; RabbitMQ async event-driven processing; 200+ PostgreSQL query optimizations (30% performance gain); Docker microservices redesign (50% scalability improvement); GitHub Actions CI/CD with automated unit, performance, and regression testing; AWS S3/EC2/RDS/Glue infrastructure.
- **Observability & engineering excellence** - built structured logging, monitoring, and alerting infrastructure; authored runbooks; led incident response with RCA; built internal debugging tools reducing triage time 40%; kept bug count below 10 across 20+ production modules over 120+ Agile sprints. Zero compliance violations across 7yr ownership - the engineering commitment to track record CoStar values.
- **Technical leadership & mentorship** - wrote design documents and architecture decisions for 20+ production modules; led projects spanning multiple engineers; mentored 5+ engineers on system design and production practices; 300+ peer code reviews; built consensus across product, compliance, and engineering stakeholders; Python/Bash automation eliminating 80% of manual workload.

Entrata India - Pune
Jan 2017 - Sep 2018

Associate Software Engineer → Trainee

- Built backend modules (Python, PostgreSQL) for production SaaS; data pipeline for 1.6M records; AWS S3 migration resolving 90% of legacy storage constraints.

PROJECTS

AgentProbe: AI-Powered Product Platform - AI-First Engineering in Practice

AI-First · Agent Workflows · CI Gate · Observability · Vibe Coding · Open Source

Python · FastAPI · LangGraph · React.js · TypeScript · ChromaDB · GitHub Actions · Anthropic API · OpenAI API | github.com/venkatalakshmiparimala/AgentProbe

Production AI-powered platform built end-to-end using AI-first engineering workflows - the development culture CoStar's greenfield platform team is building. **AI-first development:** designed, coded, tested, and shipped using Claude Code with AI-augmented workflows (code generation, test generation, debugging, architecture review) throughout - not AI as an afterthought, but as the primary development approach. **Agent-driven product features:** 15-endpoint FastAPI backend with LangGraph orchestration; multi-provider LLM routing (Anthropic, OpenAI, Ollama) with cost and latency tradeoffs; 5-metric quality engine (correctness, groundedness, hallucination rate, latency, token efficiency). **Engineering excellence:** CI regression gate (GitHub Actions) blocking quality drift before deployment; structured tracing across all agent steps; observability and monitoring dashboards. Open-source, zero regressions across 1,240+ test cases.

Leasel: AI-Powered Real Estate Workflow - Agent-Driven Product in Production

Real Estate · Multi-Agent · LangGraph · RAG · HITL · Production

Python · LangGraph (8-agent) · FastAPI · React.js · TypeScript · RAG · SQLite · MCP · Anthropic API | github.com/venkatalakshmiparimala/Lease_complaince_rag

Production AI-powered real estate workflow system - demonstrates agent-driven product features in the property domain CoStar serves. **Real estate domain:** built on 8 years of Entrata property management knowledge - lease compliance, income verification, property data workflows, regulatory requirements - directly transferable to CoStar's commercial real estate platform. **Agent-driven product:** 8-agent LangGraph StateGraph (Triage → Retrieval → Configuration → Historical → Structured Data → Reasoning → Validation → Escalation) automating complex investigation workflows with conditional routing and dynamic task delegation. **Human-in-the-loop:** deterministic HITL escalation enforces safe execution on regulated decision paths - appropriate human oversight baked into product architecture. 93.6% precision, 87.9% F1 across 1,240 test cases.

AI on FHIR: Full-Stack AI Application - AWS, React.js & TypeScript

Full Stack · AWS · React.js · TypeScript · REST APIs · Production

Python · Flask · React.js · TypeScript · Docker · AWS Lambda · PostgreSQL · REST APIs | github.com/venkatalakshmiparimala/AI_on_FHIR

Full-stack AI-powered application on AWS - React.js + TypeScript frontend with Python REST API backend; LLM-powered tool invocation translating natural-language queries into structured API calls; AWS Lambda serverless with API gateway; PostgreSQL data services; HIPAA-compliant RBAC with full audit trail and observability stack; Docker containerization. Demonstrates the full-stack engineering capability (React.js + TypeScript UI, Python backend, AWS cloud, production observability) that CoStar's greenfield platform requires - ~60% reduction in manual retrieval time.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — May 2026 · GPA 3.87 · Virginia local

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017